

Vega: Eligibility-Aware Patient-to-Trial Search

CSD358 Information Retrieval – Mid-term Hackathon, Track T6 (Vertical Search for Science and Medicine)

GitHub Repository: <https://github.com/Nithurshen/vega> · Demo Video: <https://youtu.be/vb7-pzUozck>

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Abstract

Vega matches a patient’s clinical case description to the clinical trials on ClinicalTrials.gov that the patient could join. The hard part is not finding trials about the right disease, but avoiding trials the patient is *excluded* from. Vega indexes all 375,580 trials of the TREC Clinical Trials collection with an inverted index built from scratch. Each trial is split into seven zones, and its age and sex limits are stored as parameters. Patient notes are turned into a structured profile: age and sex, expanded clinical abbreviations, negated findings and family history. That profile is matched against the trial’s structure through four lexical steps: BM25F, an age/sex eligibility test, pseudo-relevance feedback from the trials’ *conditions* zone, and an exclusion penalty. Our main novelty is a *multi-view* neural re-ranker. A zero-shot biomedical cross-encoder (MedCPT) reads the patient note against three separate views of each trial: its topic, its inclusion criteria and its exclusion criteria. Agreement with the inclusion criteria is rewarded, agreement with the exclusion criteria is penalised, and the result is fused with the structured eligibility test. Tuned only on the 75 TREC 2021 patients and tested on the 50 TREC 2022 patients with the official judgements:

- the lexical system alone reaches nDCG@10 0.533, against 0.465 for BM25F;
- Vega-Neural reaches nDCG@10 0.611 and P@10 0.494 (+31% over BM25F, $p = 1.3 \times 10^{-6}$), on par with the best of the eleven teams at TREC 2022 (0.613).

1 Problem and track relevance

User need. Most clinical trials struggle to recruit enough patients in time [16]. A clinician, or a patient, who wants to know “which trials could this person join?” faces more than 375,000 registered studies. Each has dozens of free-text eligibility criteria. A useful result list must satisfy two conditions:

- it is *about* the patient’s condition;
- the patient actually *qualifies*: their age and sex are in range, and they meet no exclusion criterion.

The TREC organisers note that real users are “extremely dissatisfied when shown trials that they are explicitly excluded from” [16].

Why Track T6. Trial records are a clear example of a structured professional domain:

- *zones*: title, summary, description, conditions, interventions, inclusion criteria and exclusion criteria;
- *parametric fields*: minimum and maximum age, sex;
- *domain language*: clinical abbreviations and negation.

Vega’s central idea is to exploit that structure, especially the inclusion/exclusion polarity, rather than matching flat text.

Benchmark and related research. We use the TREC Clinical Trials track [11, 16], which follows the earlier patient–trial test collection of Koopman and Zuccon [7]. The data:

- 375,580 trials from an April 2021 snapshot of ClinicalTrials.gov;
- 75 (2021) and 50 (2022) synthetic patient descriptions written by clinicians;
- 35,832 and 35,394 graded judgements (eligible, excluded, not relevant), pooled from participant runs.

The track itself pooled 41 runs from 11 teams in 2022 [11]. Strong systems combine BM25 first-stage retrieval with neural query generation and re-ranking [9, 10], and more recently with large language models [6]. We use MedCPT [5], a biomedical cross-encoder trained on 255 million PubMed search-log clicks, without any fine-tuning on trials. On the clinical-NLP side we build on NegEx [2] and ConText [3] for negation and experimenter (family-history) detection.

2 How we used IR

Figure 1 shows the pipeline and Table 1 maps each principle to the code. Everything on the IR side is our own NumPy code; no search library is used.

Patient profile. A note is split into clauses. A NegEx-style scan [2] marks the next six tokens after a negation cue (*no*, *denies*, *without*, *negative for*, ...) as negated, plus up to three tokens before a closing cue (... *was*

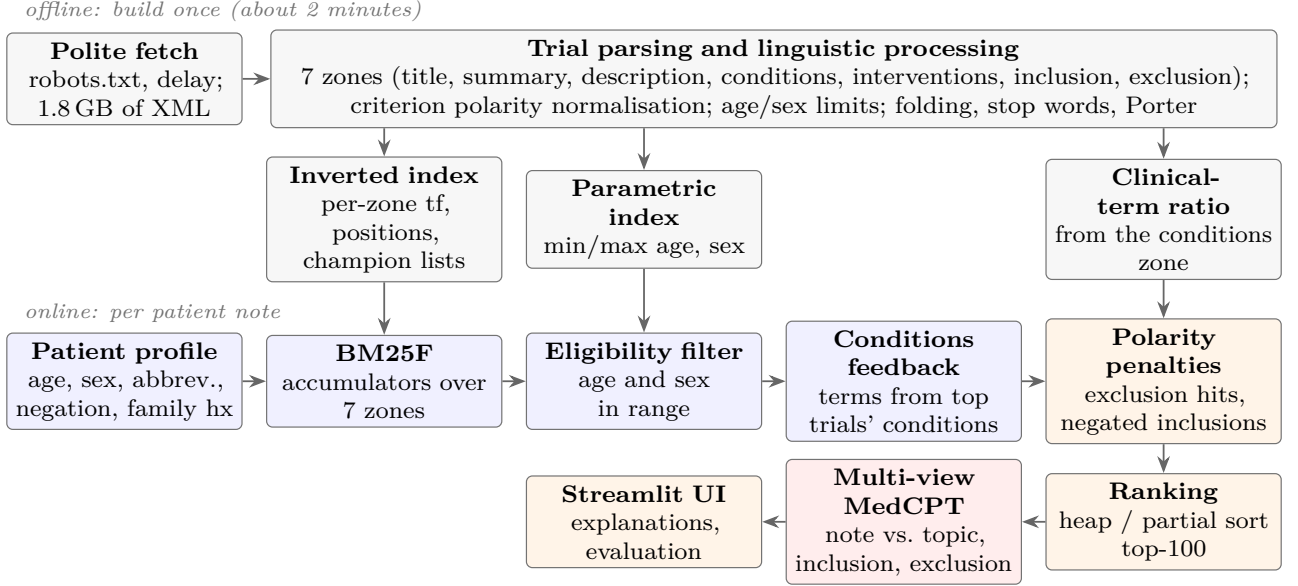


Figure 1: The Vega pipeline. Parsing 375,580 XML records and building the index take about 2 minutes on 14 cores. A lexical query takes about 122 ms; multi-view re-ranking of the top 100 adds about 5 s on a laptop GPU.

negative). Following ConText [3], clauses that mention a relative are set aside as family history. Age and sex come from patterns such as “45-year-old man” and “48 M”; we take the *earliest* age mention, so a mother’s age later in an infant’s note is ignored. Over all 125 topics this finds an age in 125, a sex in 124 (the one miss is an infant whose sex is never stated), negations in 94 and family history in 39.

Scoring. For each positive term t of the patient profile (with query frequency $qt f_t$) and each trial d :

$$\tilde{t}f_{t,d} = \sum_{z=1}^7 w_z \frac{t f_{t,d,z}}{1 - b_z + b_z |d_z|/|d_z|}, \quad s(q, d) = \sum_t \text{idf}_t \frac{(k_3 + 1) q t f_t}{k_3 + q t f_t} \cdot \frac{\tilde{t}f_{t,d} (k_1 + 1)}{\tilde{t}f_{t,d} + k_1}.$$

Zone weights were tuned on 2021 by coordinate ascent: $w = (1, 0.5, 0.5, 1, 0.5, 1, 0)$ for title, summary, description, conditions, interventions, inclusion and exclusion.

Eligibility. A trial passes when $\text{min_age} \leq \text{age} \leq \text{max_age}$ and its sex field is *All* or matches the patient. Tuning on 2021 compared a hard filter with a soft penalty (subtracting 0.5 from the normalised score) and chose the **soft** variant.

Conditions feedback. From the top 5 eligible trials we count the terms in their *conditions* zone, rank them by $\text{votes} \times \text{idf}$, and add the top 20 to the query with weight $1 \cdot \text{votes}/5$. The conditions zone is a curated, vocabulary-controlled field (it also includes MeSH terms), so feedback pulls a noisy note towards the patient’s actual diagnosis.

Polarity-aware penalties. Let $\hat{s}$ be the text score normalised by its maximum. We define the *clinical ratio* of a term as the share of its postings that fall in the conditions zone. Generic words such as *history* and *year* score 0.002, while *astrocytoma* scores 0.65. Only terms with ratio ≥ 0.05 take part in the penalties. Then

$$\text{score}(d) = \hat{s}(q, d) - \gamma \hat{s}_{\text{exc}}^+(q, d) - \delta \hat{s}_{\text{inc}}^-(q, d),$$

where:

- $\hat{s}_{\text{exc}}^+$ is BM25 restricted to the *exclusion* zone, computed over the patient’s *positive* findings that are *not* the trial’s own target condition (i.e. not in its conditions zone);
- $\hat{s}_{\text{inc}}^-$ is BM25 restricted to the *inclusion* zone, computed over findings the patient *does not have*.

Tuned weights: $\gamma = 0.2$, $\delta = 0$.

Multi-view neural re-ranking. For each of the top 100 trials, MedCPT scores the pair (patient note, view) for three views: *topic* (title, conditions and summary), *inclusion* criteria and *exclusion* criteria. Each view uses at most 384 tokens. With every score min–max normalised over the 100 candidates,

$$\text{score}'(d) = (1 - \alpha) \hat{s}(d) + \alpha c_{\text{topic}}(d) + \beta c_{\text{inc}}(d) - \gamma' c_{\text{exc}}(d) + \eta [\text{age/sex fit}].$$

Only these four weights are learned, by grid search on 2021: $\alpha = 0.5$, $\beta = 0.4$, $\gamma' = 0.1$, $\eta = 0.2$. The model itself is not trained on any trial data.

IR principle	Where	Why we chose it
Polite fetching	<code>fetch.py</code>	robots.txt is checked for both hosts (NIST and trec-cds.org), with a delay between requests and resumable downloads. The 1.8 GB of zips are streamed, never unpacked to disk.
What is a document; zones	<code>trials.py</code>	One trial is one document with seven zones. Eligibility text is split at the “Exclusion Criteria” header (present in 95.3% of trials). <i>Criterion polarity normalisation</i> moves bullets that start with <i>No</i> , <i>Not</i> , <i>Must not</i> and similar into the exclusion zone. This gives 7,700 more trials an exclusion zone, for 97.2% coverage overall.
Tokenisation, folding, stop words, Porter	<code>text.py</code>	Our own Porter stemmer, which agrees with NLTK’s original algorithm on every test word longer than two letters.
Equivalence classing (query side)	<code>patient.py</code>	70 clinical abbreviations can be expanded (HTN→hypertension, SOB→shortness of breath); ambiguous ones (MS, PE) are left out. Tuning on 2021 switched expansion <i>off</i> , because expansions add generic words such as <i>history</i> .
Inverted index, positional postings	<code>index.py</code>	357,116 terms, 61,847,191 postings with a separate tf for each of the 7 zones, and 81,462,219 positions (all zones except the long description). Built in parallel: each worker keeps a local dictionary that is then merged into one global dictionary.
Parametric index	<code>index.py</code> , <code>rank.py</code>	Age limits are converted to years and sex is coded as All/Male/Female. These arrays implement the eligibility filter and the <code>age:/sex:</code> Boolean operators.
Boolean, phrase and zone queries; skip pointers	<code>boolean.py</code>	For example, <code>conditions:"breast cancer" AND age:45 NOT exclusion:pregnant</code> . Operands are intersected in increasing-df order using $\sqrt{n}$ skip pointers.
tf-idf, BM25, BM25F	<code>rank.py</code>	Inc.ltc is the course baseline. Abstract-length notes need tf saturation, so BM25F [12] is our main text model.
Accumulators, index elimination, champion lists, top- K	<code>rank.py</code> , <code>index.py</code>	Term-at-a-time scoring into dense accumulators, with a heap for the top 10 shown in the UI and a partial sort for depth 1000. We measure the speed/quality trade-off of keeping only the top- m terms and of champion lists ($r=500$).
Relevance feedback	<code>rank.py</code>	Rocchio-style pseudo-relevance feedback [8, 13] restricted to the <i>conditions</i> zone of the top trials.
Two-stage retrieval, re-ranking	<code>rank.py</code> , <code>neural.py</code>	The cheap lexical index selects the top 100; an expensive cross-encoder re-ranks only those. Recall@100 of the first stage caps what re-ranking can fix.
Evaluation	<code>evaluate.py</code>	nDCG@10, P@10, RR, R@1000, the share of excluded trials in the top 10, judged@10 and condensed-list scores [14].

Table 1: IR concepts used in Vega, where they live, and why.

3 Beyond IR

- **Clinical NLP.** We reimplemented the core of NegEx and ConText (negation scope and experienter) as lightweight rules. Regex-based extraction of age and sex covers notes, unit-bearing age limits (“6 Months”) and de-identification markers.
- **Neural retrieval.** MedCPT is a PubMedBERT cross-encoder. It runs on the laptop’s Apple GPU (PyTorch MPS) at about 60 pairs per second, so re-ranking 100 trials in three views takes about 5 s per patient. Scores are cached per (note, trial), so tuning the fusion weights needs no extra model calls.
- **Systems.** The XML zips are streamed into a 1.5 GB JSON-lines store with a byte-offset table for $O(1)$ random access in the UI. The index is built with 12 worker processes that each keep a local dictionary, which are then merged.
- **Statistics.** We use paired t - and Wilcoxon tests over topics [15]. Because the judgement pools come from *other* systems’ runs, we also report judged@10 and condensed-list nDCG@10’ [1, 14], so that a new system is not penalised merely for retrieving unjudged trials.

4 Novelty

1. **Polarity-aware, multi-view neural matching.** Neural re-rankers normally read a document as one block of text, so “patient *has* diabetes” looks equally relevant to “*requires* diabetes” and “*excludes* diabetes”. Vega asks the same zero-shot model three separate questions: is this trial about the patient (topic)? does the

System	nDCG@10	P@10	P@10 _{≥1}	RR	R@1000	Excl@10 ↓
TF-IDF Inc.ltc (baseline)	0.462	0.342	0.556	0.539	0.598	0.214
BM25 (raw note)	0.436	0.296	0.524	0.565	0.519	0.228
BM25F zones (raw note)	0.465	0.324	0.556	0.586	0.593	0.232
+ note processing	0.457	0.316	0.534	0.595	0.559	0.218
+ age/sex filter	0.510	0.382	0.558	0.693	0.593	0.176
+ conditions feedback	0.532	0.398	0.590	0.704	0.653	0.192
Vega (full)	0.533	0.400	0.586	0.714	0.650	0.186
MedCPT alone (topic view)	0.560	0.436	0.622	0.693	0.650	0.186
Vega + MedCPT topic view	0.591	0.468	0.660	0.716	0.650	0.192
+ inclusion view	0.603	0.480	0.672	0.715	0.650	0.192
+ exclusion view	0.608	0.488	0.674	0.727	0.650	0.186
Vega-Neural (full)	0.611	0.494	0.676	0.737	0.650	0.182

Table 2: TREC CT 2022 test results (50 patients). Top block: the lexical system, where each row adds one component and Vega (full) adds the exclusion penalty (tuning set the negation penalty to 0). Bottom block: MedCPT on its own, then multi-view re-ranking of Vega’s top 100, adding one view at a time. Bold marks the best value per column; for Excl@10, lower is better.

patient match what it *requires* (inclusion)? does the patient match what it *rules out* (exclusion)? It then combines the answers with signed weights and the structured age/sex test. Section 5 shows that each view adds to the one before. We first tried this polarity idea with bag-of-words penalties; it barely helped, and the neural views are what make it work.

2. **Criterion polarity normalisation.** Many older and NCI-style records state exclusions as “No prior chemotherapy” inside a single criteria block. Moving such bullets into the exclusion zone repairs the document structure that both the lexical penalty and the neural exclusion view depend on.
3. **The conditions zone as free supervision.** The curated conditions zone serves twice: as a data-driven classifier of clinical vs. generic words (the clinical ratio), and as a controlled vocabulary for feedback.
4. **An honest evaluation under pooling.** Tuning is on 2021 only, results are on 2022, and we report both official and condensed metrics plus judged@10.

5 Evaluation

Setup. We use the official TREC protocol [11, 16]. nDCG@10 uses gain 2 for eligible and 1 for excluded trials [4]. P@10 and RR count only eligible trials as relevant. We also report P@10 counting eligible or excluded trials (P@10_{≥1}), the share of top-10 trials that are judged *excluded* (Excl@10, lower is better) and judged@10. Every hyperparameter was tuned on the 75 patients of 2021. All numbers below are on the 50 patients of 2022, whose 35,394 judgements include 3,939 eligible and 3,036 excluded trials.

Main result (Table 2, Fig. 2).

- *Baselines.* The course baseline Inc.ltc (0.462) and BM25F on the raw note (0.465) are close. Plain BM25 over the whole record is weaker (0.436), because long description fields dominate its length normalisation.
- *Lexical Vega: structure pays.* The age/sex eligibility test gives the largest lexical gain (0.457 → 0.510 nDCG@10, Excl@10 0.218 → 0.176), and conditions feedback adds more (0.532). Lexical Vega reaches 0.533 (+15% over BM25F; paired t : $p = 0.002$).
- *Multi-view re-ranking, our main result.*
 - MedCPT on its own, reading only the topic view of Vega’s top 100, scores 0.560. Fused with the lexical score it scores 0.591, so the two signals complement each other.
 - Adding the patient-vs-*inclusion* view raises this to 0.603, the signed patient-vs-*exclusion* view to 0.608, and the age/sex term to **0.611**.
- *Vega-Neural against the baselines.* nDCG@10 0.611, P@10 0.494 and RR 0.737: +31% nDCG@10 and +52% P@10 over BM25F.
 - Against BM25F: $p = 1.3 \times 10^{-6}$ (paired t) and $p = 1.2 \times 10^{-6}$ (Wilcoxon); better for 40 patients, worse for 8.
 - Against lexical Vega: $p = 3.8 \times 10^{-4}$; better for 38, worse for 9.
- *Robustness.* Only four weights are learned, all on 2021. Selecting them by 5-fold cross-validation *within* 2021

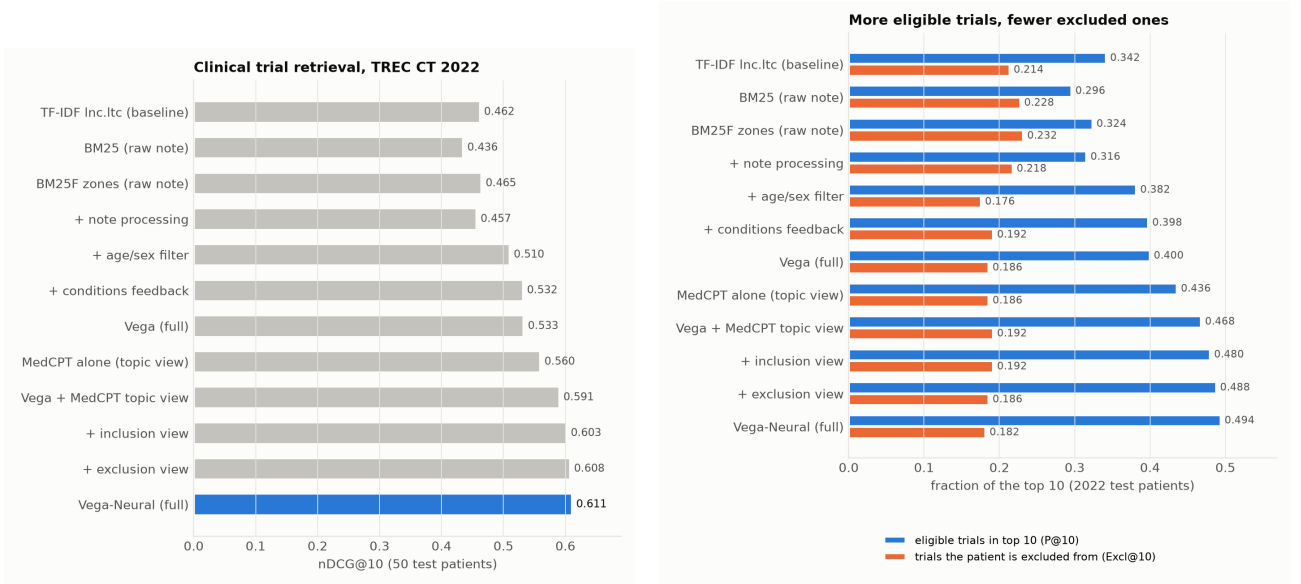


Figure 2: Left: nDCG@10 per system. Right: eligible trials (P@10) and trials the patient is excluded from (Excl@10) in the top 10.

System	nDCG@10	P@10	P@10 _{≥1}	RR	R@1000	Excl@10 ↓
Vega (full)	0.533	0.400	0.586	0.714	0.650	0.186
- note processing	0.550	0.418	0.610	0.717	0.687	0.192
- age/sex filter	0.484	0.346	0.556	0.610	0.615	0.210
- conditions feedback	0.505	0.382	0.550	0.693	0.587	0.168
- exclusion penalty	0.532	0.398	0.590	0.704	0.653	0.192
hard filter instead	0.545	0.412	0.604	0.719	0.667	0.192

Table 3: Ablations on 2022: each component removed from Vega.

Setting	postings scanned	lexical ms	nDCG@10
index elimination, m=10	24,866	2	0.368
index elimination, m=20	110,694	6	0.443
index elimination, m=40	1,047,095	36	0.459
index elimination, m=80	2,532,673	79	0.465
index elimination, all	2,532,673	79	0.465
champion lists, r=500	24,085	14	0.369

Table 4: Index elimination and champion lists (BM25F, 2022).

also improves held-out folds (0.553 → 0.595), so the gain is not tuned to one year.

- *Pooling check.* judged@10 is 0.958 for Vega-Neural, against 0.970 for BM25F, and the condensed-list score agrees (nDCG@10' 0.622 vs. 0.471). The gain is not an artefact of unjudged trials.
- *Context.* The best runs of the top six of the 11 teams at TREC 2022 scored nDCG@10 0.613, 0.557, 0.505 (manual), 0.491, 0.476 and 0.453, and P@10 0.508, 0.456, 0.398, 0.362, 0.354 and 0.322 [11]. Vega-Neural's 0.611 and 0.494 are **on par with the best team** and ahead of all the others. It achieves this with a model that was never trained on trials and four learned weights. Our runs were not in the official pools, which if anything works against us.

Ablations (Table 3, Fig. 3).

- *What matters most.* Removing the age/sex test costs the most (nDCG@10 0.484, -0.050), followed by conditions feedback (0.505, -0.028).
- *What the views do.* The neural gain is mainly *topical*: P@10_{≥1} (eligible or excluded) rises the most. The share of excluded trials in the top 10 moves only slightly (0.192 with the topic view, 0.182 with all views). The eligibility-specific views help, but modestly, because whole criteria zones are truncated at 384 tokens. Criterion-level matching is therefore first on the roadmap.
- *Lexical penalties.* The lexical exclusion penalty is small: it helped on 2021 (0.551 → 0.553) and changes 2022 nDCG@10 by -0.001 when removed, while lowering Excl@10. A zone-level bag of words is too coarse to decide eligibility, which is why criterion-level matching heads the roadmap. Tuning gave the negation penalty weight 0: findings a patient lacks are rarely written verbatim in inclusion criteria.
- *Two 2021 choices did not transfer.* 2021 tuning selected negation and family-history removal (without abbreviation expansion) and a soft age/sex penalty. On 2022, raw-note text would reach 0.550 and a hard filter 0.545. With 50–75 patients, differences under 0.02 are within topic-to-topic variance, so we report the system chosen on 2021 rather than tuning on the test set.
- *Efficiency* (Table 4). Index elimination to the top $m=40$ query terms halves lexical time (79 → 36 ms) and scans $2.4\times$ fewer postings, at a small cost (0.459 vs. 0.465). Champion lists are faster still (14 ms) but fall to 0.369: a patient note's evidence is spread over many moderately rare terms, and few trials appear in any single term's champion list.
- *Stemming.* Porter shrinks the vocabulary by 15% and lifts nDCG@10 from 0.516 to 0.533 ($p = 0.13$, not

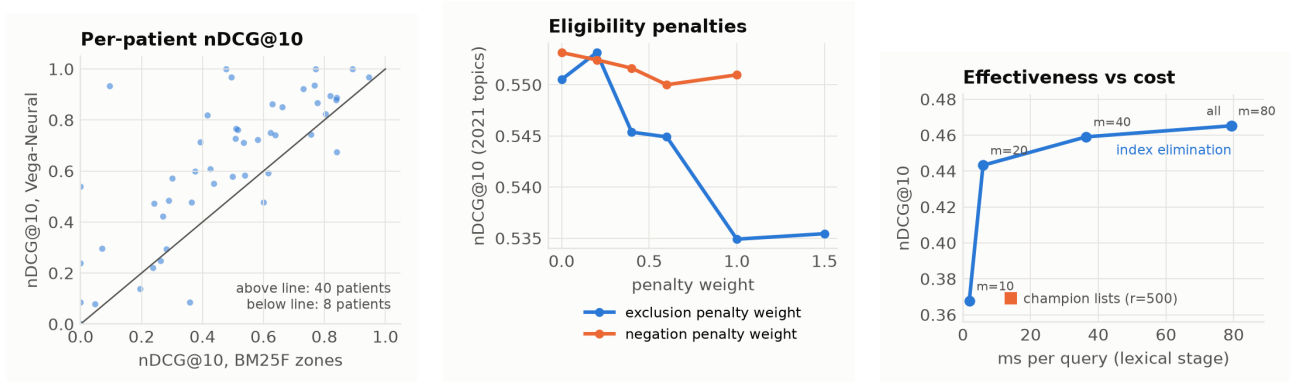


Figure 3: Left: nDCG@10 for each 2022 patient, Vega-Neural against BM25F. Middle: 2021 tuning curves for the two penalties. Right: efficiency on 2022.

significant).

6 Limitations and next steps

- **Pooling bias.** Vega retrieves trials that no 2022 participant returned. These count as non-relevant in the official score; judged@10 and the condensed metric show how large the effect is.
- **Whole-zone criteria.** Both the lexical penalty and the neural views read a whole criteria zone, not individual criteria. Long criteria lists are cut at 384 tokens, and lab thresholds (“HbA1c > 9%”) and time windows (*within 6 months*) are not reasoned about.
- **Cost.** Neural re-ranking adds about 5 s per patient on a laptop GPU, and it cannot recover relevant trials that the lexical stage left out of the top 100.
- **Rule-based note processing.** The abbreviation list is hand-built, and negation and family history are decided per clause. In test patient 2022-7, the clause “...achondroplasia in both father and the child” mentions a relative, so the patient’s *own* diagnosis was set aside as family history. Feedback then drifted to FGFR3 cancer trials: lexical nDCG@10 is 0.07, against 0.55 with family-history removal switched off. The neural re-ranker, which reads the full note, recovers to 0.485. Experiencer detection still needs to work on noun phrases, not whole clauses.
- **No concept normalisation.** UMLS or MeSH mapping (e.g. *heart attack* $\equiv$ *myocardial infarction*) would close many vocabulary gaps.

7 Work division

This is a solo submission, approved by the course instructor.

Member	Components
K S Nithurshen (2410110157)	Topic and dataset selection, design, data fetching, trial parsing, patient-note processing, indexing, Boolean search, ranking and eligibility logic, evaluation, Streamlit UI, report and demo video.

8 AI-use declaration

Used **Claude** (Anthropic, model Claude Opus 5.5) for the report and video demonstration script assistance only.

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A Walk-through of one test patient

Patient 2022-5 (test set): *A 23-year-old man comes to the emergency department following an episode of syncope. He was working out when he felt dizzy and passed out without head injury. He has had 3 other episodes of light-headedness over the last year, all happening during physical activity. He never had this experience while resting. He has no other medical conditions. The patient does not use tobacco, alcohol, or illicit drugs. His father died suddenly at age 35. Vital signs are within normal limits. On physical examination, the patient has a harsh systolic murmur. The lungs are clear with no peripheral edema. Echocardiography shows asymmetric interventricular septal hypertrophy.*

Vega’s profile: age 23, sex male; abbreviations none; negated findings: head, injury, experience, resting, medical, conditions.

#	BM25F zones (raw note)	Vega-Neural
1	The Cardiopulmonary Effect of Inhaled Beta-2-... (unjudged)	The Role of Heart Stiff and Weak Atrium on Ex... (eligible)
2	The Role of Heart Stiff and Weak Atrium on Ex... (eligible)	Role of Perhexiline in Hypertrophic Cardiomyo... (eligible)
3	Copeptin and the S-100b Protein in Stroke (not rel.)	Study of Hypertrophic Cardiomyopathy Under St... (eligible)
4	Mechanisms of Vasovagal Syncope (excluded)	Speckle Tracking Echocardiography Analysis of... (eligible)
5	Alcohol Septal Ablation in Obstructive Hypert... (eligible)	Study to Develop a Non-invasive Marker for Mo... (eligible)

B Final hyperparameters (tuned on 2021)

- BM25: $k_1 = 1.2$, $b = 0.75$, $k_3 = 8$.
- BM25F zone b : $(0.5, 0.75, 0.75, 0.5, 0.5, 0.75, 0.75)$; zone weights $w = (1, 0.5, 0.5, 1, 0.5, 1, 0)$.
- Note processing: negation removal, family-history removal. Negation scope: 6 tokens forward, 3 tokens backward for closing cues.
- Eligibility: soft (penalty 0.5 for an age/sex mismatch).
- Conditions feedback: 5 trials, 20 terms, weight 1.
- Penalties: $\gamma = 0.2$, $\delta = 0$, clinical ratio ≥ 0.05 .
- Neural re-ranking: top 100, MedCPT cross-encoder, 384 tokens per view, $\alpha = 0.5$, $\beta = 0.4$, $\gamma' = 0.1$, $\eta = 0.2$.